confirmed that a spinning black hole must radiate and slow its spin. However, they also predicted that, when the hole stops spinning, its radiation does *not* stop. With no spin left, and no spin energy left, the hole keeps on emitting radiation of all sorts (gravitational, electromagnetic, neutrino), and as it emits, it keeps on losing energy. Whereas the spin energy was stored in the swirl of space outside the horizon, the energy now being lost could come from only one place: from the hole's interior!

Equally amazing, Hawking's calculations predicted that the spectrum of the radiation (that is, the amount of energy radiated at each wavelength) is precisely like the spectrum of thermal radiation from a hot body. In other words, a black hole behaves precisely as though its horizon has a finite temperature, and that temperature, Hawking concluded, is proportional to the hole's surface gravity. This (if Hawking was right) was incontrovertible proof that the Bardeen—Carter—Hawking laws of black-hole mechanics *are* the laws of thermodynamics in disguise, and that, as Bekenstein had claimed two years earlier, a black hole has an entropy proportional to its surface area.

Hawking's calculations said more. Once the hole's spin has slowed, its entropy and the area of its horizon are proportional to its mass squared, while its temperature and surface gravity are proportional to its mass divided by its area, which means inversely proportional to its mass. Therefore, as the hole continues to emit radiation, converting mass into outflowing energy, its mass goes down, its entropy and area go down, and its temperature and surface gravity go up. The hole shrinks and becomes hotter. In effect, the hole is evaporating.

A hole that has recently formed by stellar implosion (and that thus has a mass larger than about 2 Suns) has a very low temperature: less than 3×10^{-8} degree above absolute zero (0.03 microkelvin). Therefore, the evaporation at first is very slow; so slow that the hole will require longer than 10^{67} years (10^{57} times the present age of the Universe) to shrink appreciably. However, as the hole shrinks and heats up, it will radiate more strongly and its evaporation will quicken. Finally, when the hole's mass has been reduced to somewhere between a thousand tons and 100 million tons (we are not sure where), and its horizon has shrunk to a fraction the size of an atomic nucleus, the hole will be so extremely hot (between a trillion and